

Faisal D'Souza
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Networking and Information Technology Research and Development Program
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Dear Mr. D'Souza:

Microsoft appreciates the opportunity to provide comments to the White House Office of Science and Technology Policy (OSTP) in its request for information for the Development of an Artificial Intelligence (AI) Action Plan. As a leading U.S. technology company deeply committed to sustaining and enhancing America's leadership in AI, we want to first acknowledge and thank President Trump for his strong leadership on AI during his first term. As our Vice Chair and President, Brad Smith, recently [wrote](#), we believe this is a golden opportunity for American AI.

We share the Administration's goal: the United States must stay at the forefront of AI. At Microsoft, we have a responsibility to enable innovation and foster competition in this dynamic AI ecosystem. AI is a foundational technology with a transformative capability to help solve societal problems, improve human productivity, and make companies and countries more competitive. That is why Microsoft is investing over \$50 billion in America's AI infrastructure in 2025, with projects in places like Wisconsin and Arizona creating state-of-the-art datacenters and thousands of jobs. Our investment in America's AI infrastructure underscores the crucial role American companies play in advancing local economic growth by providing the infrastructure communities need to power this digital transformation.

As we promote innovation and adoption of these powerful AI tools, we must take steps to build trust in AI, ensure we protect the public from abusive AI, and defend our national security. As AI technology rapidly evolves, it is crucial to safeguard our nation's critical infrastructure and fortify our AI labs from malicious actors. Ensuring that our laws are updated to protect people from AI-enhanced fraud and non-consensual imagery is critical. We applaud the First Lady's energy and leadership on the latter.

Most importantly, we offer ourselves as a resource to the Administration and propose several policy recommendations across three critical topics for your consideration: 1) Accelerating American AI, 2) Protecting the Public and Enhancing National Security; (3) Investing in Scientific Research and Science-based AI Standards.

Accelerating American AI

Today, the United States is the world-leader in cutting-edge AI infrastructure, frontier model development, and training a skilled workforce to power a robust AI driven economy. China

is investing in AI innovation to try to catch up and outcompete the United States. Continued innovation and leadership by the United States at a global scale will require that the private sector and government coordinate and work together to harness the full potential of our domestic capabilities, like advanced semiconductors, world-class engineering talent, and access to low-cost power. It will also be driven by the ongoing partnerships between companies that operate at various levels of the AI technology stack—from chip designers to hyperscalers to AI model developers. Much like the advent of electricity, AI is a general-purpose technology. AI holds the potential to revolutionize industries, enhance productivity, and create entirely new business categories. We need to recognize where we are at this moment in time with the advent of powerful AI and its long-term importance to people and society by investing in AI infrastructure, AI skilling, and AI data to keep our lead.

Power the Growth of AI Infrastructure

The new AI economy will require enormous computational power. Historically, such power could only be found in a handful of government-funded institutions, and only available to a select few. But the advent of the public cloud changed that. Today, datacenters around the world host millions of servers and make vast computing power broadly available to organizations large and small. New datacenter investments, like those Microsoft is making across the United States, will aid tech startups, enterprises, government agencies, research labs, and non-profit organizations around the country as they innovate new AI tools and businesses.

Datacenters play a vital role as the economic engines of communities and states. They support mission-critical operations in a 24/7 digital environment, drive economic growth in regions, and improve productivity across sectors and industries. But they also require electricity in large sums to power this digital economy, as well as critical inputs to hook up to the grid, like transformers. To power this infrastructure, the United States will need to invest in more energy resources, onshore our supply chains, and modernize our electric grid to support forecasted increases in electrical loads.

As the United States sees more adoption of cloud services, AI, and the reshoring of manufacturing, the need for abundant, affordable, and reliable power will continue to accelerate. This growing demand underscores the importance of timely investments in new generation, transmission, and distribution infrastructure to support economic growth and national security. To support the growing demand for electricity, ensure grid reliability, and connect new generation sources, it is essential to modernize and expand the existing grid infrastructure. Our advanced economy relies on 50-year-old infrastructure that cannot meet the increasing electricity demands driven by AI, reshoring of manufacturing, and increased electrification.

There are several areas where the federal government can fuel innovation, support the development of energy infrastructure, and accelerate the permitting process. Therefore, we recommend the federal government:

- **Streamline the federal permitting process to accelerate economic growth.** The federal government should take immediate steps to establish reliable, reasonable, and transparent timelines for permitting decisions. Furthermore, the federal government could provide standardization for federal permits related to datacenters to ensure consistent timelines across permitting agencies in various parts of the country. The federal government should enable the use of AI and advanced digital technologies to accelerate decision-making.
- **Increase resources in energy development.** It is critical that energy-generation resources be added to the grid, including nuclear and fusion, at a pace to keep up with demand. For example, Microsoft has signed a 20-year power purchase agreement with Constellation to restart the Three Mile Island Unit 1, now renamed the Crane Clean Energy Center, which will add 835 megawatts of carbon-free electricity to the grid. This will create 3,400 direct and indirect jobs, while also contributing over \$3 billion in local, state, and federal taxes. To accelerate energy development, it is essential to improve the interconnection process and provide faster determinations. Furthermore, the federal government should increase research and development dollars as well utilize tax incentives to drive accelerated development.
- **Advance the Federal-State Modern Grid Deployment Initiative.** Help enhance grid technologies with federal support by modernizing the Federal Energy Regulatory Commission (FERC)'s authority for interregional transmission planning to boost capacity, reduce congestion, and ensure efficient permitting for new transmission lines.
- **Increase domestic manufacturing of energy infrastructure.** There is a need for additional manufacturing capacity for electricity grid infrastructure. Using economic tools, like tax incentives, and ensuring adequate supply of key grid infrastructure components as the U.S. builds increased capacity will reduce our dependence on foreign sources for electric grid infrastructure.
- **Enhance resilience of AI supply chain.** U.S. capacity is limited not only in critical minerals, but also key components like printed circuit boards, generators/generator batteries, cooling liquids and systems, and networking equipment. Investment in this area is necessary to address these gaps. Cooperation with allies and partners, particularly where there is a lack of domestic access or low margins, makes production impractical or unattractive for reshoring.

Skill the American Workforce for the AI Future

AI has the potential to bolster the productivity and quality of work for American workers. We also expect advances in AI to create millions of new jobs over the next decade including novel kinds of jobs that do not exist today. To keep America competitive and ensure Americans of all backgrounds have the critical AI skills necessary to benefit from these opportunities, the government should take a holistic workforce development approach to ensure every American can participate in the AI economy. Most individuals will require basic AI literacy skills, enabling them to leverage existing technology and streamline other

functions in the workplace. Some individuals will need deeper training that builds upon existing disciplines like applied science, cybersecurity, machine learning, security, and data science, potentially evolving into a new generation of AI engineering. Others will require interdisciplinary expertise and skills to design or manage the integration of AI systems into the business processes that support organizations across the private, public, and non-profit sectors. In some instances, individuals will need to learn skilled trades to build and maintain data centers for AI. To meet the totality of these skill needs, the country must pursue a new national goal to make AI skilling accessible and useful for every American. This will require a broad range of partnerships and public policies, spanning geographic, organizational, and economic divides.

In furtherance of President Trump's 2019 EO 13859, Maintaining American Leadership in Artificial Intelligence, which raised the importance of training current and future generations of American workers with skills for today's economy and jobs of the future, we recommend the federal government:

- **Target federal apprenticeships.** Target current federal apprenticeship investments into regions which have identified major AI infrastructure initiatives and could benefit from expanded apprenticeship opportunities to meet those building needs.
- **Implement skills-based hiring.** Accelerate efforts to implement skills-based hiring within the federal government and help share best practices with private sector employers, including technology-based jobs.
- **Enhance labor market insights.** Modernize government labor market datasets to capture the speed of change in the market, identify high-demand occupations, and focus on the emerging skillsets needed in the future. Enrich or supplement these data assets with aggregated private sector data sources, which may offer more granular, timely, and flexible data points.
- **Establish federal programs for on-the-job training.** Raising awareness to both employers and employees on the benefits of utilizing Section 127, within the Internal Revenue Code, could allow an increase in providing AI skilling. Consider new incentives to encourage employers to provide on-the-job AI skills training.
- **Reauthorize the Workforce Innovation and Opportunity Act (WIOA).** Encourage Congress to reauthorize WIOA, a U.S. law that enhances federal workforce development programs by working with states and local governments to help job seekers find employment and training, to ensure that more funding is directed to training and that programs can meet the needs of the AI and quantum future, including through employer-driven training and ensuring that training programs are relevant to the needs of today's employers and are focused on new technology.

K-12 education must adapt to the future realities of AI usage in both daily life and at work. This requires not only promoting efforts to help students understand the safe adoption of AI but also enhancing their fundamental critical thinking skills. Multiple studies indicate that investments in computer science education significantly boost students' problem-solving

and critical thinking capabilities. According to Code.org, 67% of new STEM jobs are in computing, yet only 11% of STEM graduates specialize in computer science. By incorporating AI literacy and technical skills into K-12 education, we can better prepare future workers to navigate and utilize advanced technologies, thereby driving productivity and economic growth. Therefore, we recommend the federal government:

- **Support professional development.** Provide professional development for teachers to effectively teach AI-related subjects and integrate AI tools into the classroom.
- **Support ongoing education campaigns.** Continue and build upon current efforts to educate schools and school districts and provide technical assistance to help them responsibly deploy AI.
- **Support public-private partnerships.** Incentivize and prioritize federal funds and grants that support partnerships between technology companies and educational programs, including apprenticeship and credentialed organizations, to develop up-to-date AI curriculum.

Ensure U.S. Innovators Have Access to High Quality Data for AI

To keep the United States at the vanguard of global AI development, we need a healthy AI data ecosystem full of vast data to analyze themes, patterns, and insights across broad datasets. Aside from the world wide web, the federal government has the largest source of high-quality and high-volume data, such as in the National Archives, the Library of Congress, the Smithsonian, the Department of Commerce, National Institute of Health, and across many government agencies. These institutions lack the resources to make data accessible for AI training. Much of the data is stored in non-digital formats, and even when it is digital, it may be offline or locked behind paywalls in closed-access journals, restricting access to some of the most valuable scientific knowledge in the world. Making this knowledge available for AI training can advance the capabilities of AI models, driving innovation and discovery.

In comparison, China is already investing in data resources with a plan to build a comprehensive system to manage datasets as a strategic resource by 2030. Its AI development is built on a robust foundation of vast data resources, government initiatives, and integrated data collection infrastructure, positioning it to lead in the global AI race. China is not alone in recognizing the economic and strategic value of national-scale data management. The United Kingdom is developing a National Data Library to foster its AI sector, highlighting a growing global trend of treating data as common for economic competitiveness. Therefore, we recommend the federal government:

- **Support a national open data campaign.** Launch an initiative to spur discoverable, accessible, and usable data for U.S. competitiveness and innovation, by responsibly providing access to non-classified, non-sensitive state and federal government data.

- **Promote the creation and expansion of open data commons.** Support and encourage non-profit, academic, and cultural institutions to make valuable data more discoverable, accessible, and usable through the establishment of open data commons.
- **Remove restrictions and paywalls.** Create a fund to open scientific research currently restricted by closed-access journals and paywalls for AI training and advancement.
- **Formulate and update incentives for data sharing.** A great deal of federally funded research comes with a requirement that scientists openly share data collected as part of research. However, in many cases data is not shared as required in funding agreements. Stronger incentives are needed to ensure that data is openly shared in a secure and timely manner, via the contracts made with funding agencies.
- **Support the development and use of data annotations standards.** Support the development and use of standards for clearly annotating datasets with metadata that clearly describe the source, veracity, and provenance of datasets. Tracking the provenance of data is especially important given the growth of synthetic data, created by generative AI models.

To support AI innovation, our ambition for data must equal our ambition for infrastructure and talent. Without it, AI systems will stagnate. This includes access to data contained within online content, acquired data, synthetic data, and proprietary data. Open and fair access to data at scale—including copyrighted data—is critical for AI training. Efforts to undermine this goal will thwart AI progress. Fortunately, our existing “fair use” copyright laws, support advancing economic growth and prosperity.

Grounded in the Constitution and designed to promote innovation, but not lock-up information, ideas and data, U.S. copyright law serves the dual function of protecting creative works and innovation. The fair use doctrine permits the use of copyrighted material without the permission of the rightsholder for certain purposes such as education, research and innovative applications that do not interfere with exploitation of the original work. Notwithstanding pending federal court cases examining this issue, it would grind U.S. AI innovation to a halt if the training of models on copyrighted works is not considered fair use. As tech investors like Andreessen Horowitz have noted in their response to the Copyright Office’s Notice of Inquiry, “There is a very real risk that the overzealous enforcement of copyright when it comes to AI training—or the ad hoc limitation of the fair use doctrine that properly protects AI training—could cost the United States the battle for global AI dominance.” We could not agree more. To avoid this outcome, we recommend the federal government:

- **Enforce current law.** Enforce existing copyright law and uphold the fair use doctrine, while also not allowing other countries to impose their version of copyright law onto our companies.

Protecting the Public and Enhancing National Security

AI's integration into our economy and society is reshaping how nations protect their citizens, secure borders and critical infrastructure, and maintain global competitiveness. In doing so, we must also ensure consumers trust the technology. Trust is the foundation of adoption. By ensuring that AI systems are secure and reliable, we can accelerate their integration into our economy, driving innovation and competitiveness, while also ensuring our national security.

AI enhances cybersecurity, intelligence analysis, and operational efficiency. AI-powered cybersecurity solutions are vital for detecting and responding to cyber threats and safeguarding critical infrastructure. AI also improves intelligence operations by analyzing vast amounts of data, enabling accurate threat assessments and decision-making. By prioritizing AI development and fostering public-private partnerships, the United States can enhance our military prowess, economic strength, and diplomatic influence.

As AI advances, its applications in national security will expand, presenting new opportunities and challenges. Reliable AI development and deployment are crucial to enhancing national security without compromising ethical standards or civil liberties. Access to both industry and open-source models is critical for government and businesses to have the latest technology at optimal cost, performance, and security. By fostering collaboration between government, industry, and academia, nations can drive technological advancements that bolster security and economic prosperity, ensuring a stable and secure future in an AI-driven world.

Protecting the Public from Abusive AI to Build Trust

The emergence of AI generated synthetic content — text, images, and videos — presents new challenges for policymakers. Synthetic content has the potential to revolutionize creative industries and enable new forms of artistic expression and innovation. However, it also harbors risks, as it can be manipulated for malicious purposes in the hands of bad actors. It will be critical for society to combat abusive AI-generated content in a [comprehensive](#) but targeted manner. Last year, in an effort to provide ideas to maintain trust in our online ecosystem, Microsoft, published a [white paper](#) setting forth recommendations for U.S. policymakers to consider in protecting children and seniors, while also encouraging AI adoption.

Additionally, we must ensure that personal information of consumers is protected in a responsible way. Microsoft has advocated for a comprehensive federal privacy law since 2005 because we know that consumer trust is critical to the adoption of new digital technologies. Therefore, we recommend the federal government:

- **Support the creation of a new “deepfake fraud statute.”** Modeled after the Truth in Caller ID Act, enact legislation that provides law enforcement officials, including

state attorneys general and the Federal Trade Commission, authority to address AI-generated fraud as they proliferate in speed and complexity.

- **Update laws.** Ensure that our federal and state laws on child sexual exploitation and abuse and nonconsensual intimate imagery are updated to include AI-generated content. We urge Congress to pass and the President to sign the TAKE IT DOWN Act into law. We applaud the Senate for passing the bill by unanimous consent and for the First Lady's endorsement.
- **Promote use of state-of-the-art provenance technologies for authentic content.** Provenance technology such as the [C2PA specification](#) was originally developed to authenticate real media and help the public better differentiate between authentic and synthetic content. The federal government can play an important role in adopting these tools and should use this technology to assist the public in identifying authoritative government outputs as authentic.
- **Support federal privacy legislation.** To ensure that consumers are protected and have full confidence in new AI systems, it is critical that Congress pass a comprehensive federal privacy law. This law should create a consistent national standard, grant consumer rights over their own information, and require responsible use of personal information while still allowing for innovation and development of new products and services.

[Harness the Power of Highly Capable AI Models for National Security](#)

The federal government should fully harness the power of highly capable American AI models to advance U.S. national security. AI models can provide groundbreaking capabilities to the U.S. defense and intelligence community. In particular, the use of AI systems powered by highly capable models can process and analyze vast amounts of data at a rapid pace to provide insights and threat detection to inform decision-making. These models can be used to enhance the efficiency of core agency functions, streamline operations, and reduce response times by providing more efficient solutions to complex challenges. The national security community has been slow to adopt highly capable AI models and use case development has been minimal. Cyber adversaries are adopting generative AI technology to gain strategic advantage with its power, reach, and capabilities and the U.S. cannot afford to fall behind the innovation curve. We encourage the federal government to consider the situations for which this powerful technology can be most effectively harnessed. Therefore, we recommend the federal government:

- **Prioritize the adoption and use of domestic AI models by U.S. defense and intelligence agencies for mission enablement and efficiency.** U.S. Cloud Service Providers (CSPs) have invested in and created a classified cloud infrastructure to allow the Defense Department and Intelligence Community to leverage the most advanced AI models at scale.

[Secure the AI Technology Stack at Home and Abroad](#)

Securing the technology stack that powers AI is crucial for maintaining American economic competitiveness, technological advantage, and national security. Beyond traditional

cybersecurity measures, this objective requires securing data centers, which house the infrastructure and platforms that host, train, fine-tune, and utilize models for inference. Across the AI tech stack, Microsoft employs state-of-the-art security controls and complies with U.S. laws and security requirements, as well as international agreements. The federal government can enhance technology stack security, support the development of robust cybersecurity measures, and promote secure development practices. Therefore, we recommend the federal government:

- **Develop comprehensive policies.** The U.S. government should work with technology companies to develop comprehensive policies that cover AI security for each layer of the technology stack. This includes the infrastructure, data, model, application, and governance layers.
- **Enhance physical and cybersecurity measures.** Implementing clear standards and guidelines for physical and cybersecurity measures for data centers and AI infrastructure.
- **Promote secure development practices.** Encourage the adoption of secure development practices for relevant AI-related code. This includes developer training, threat modeling, and using approved cryptography standards.
- **Foster collaboration and innovation.** Build connections across the technology ecosystem to share best practices, advance innovation on future-proofing security posture, and address shared challenges through regular meetings and collaboration opportunities.

Leverage AI for Enhanced Cybersecurity Protection

Harnessing AI's capabilities can significantly enhance cyber defenses, streamline threat analysis, and fortify software systems. However, AI also introduces new risks that necessitate vigilant oversight and proactive measures. Collaborative efforts between government and industry, both domestically and internationally, are crucial to safeguarding AI infrastructure and preventing its malicious use. By leveraging existing standards and fostering innovation, we can ensure that AI strengthens our defenses.

In response to the evolving digital threat landscape, where last year, we detected over 600 million cyber-attacks a day against our customers, Microsoft announced its Secure Future Initiative (SFI). SFI incorporates engineering investments and governance processes and procedures to ensure security is embedded in all our products, services, and culture. Using AI to defend against cyber adversaries is one pillar of SFI as it will provide defenders with an asymmetric advantage over the attackers. SFI also aims to protect AI systems from emerging threats by leveraging existing risk-based cybersecurity standards, such as the NIST Cybersecurity Framework (CSF) and the Secure Software Development Framework (SSDF). SFI underscores Microsoft's resolve to ensure secure design, development, and operations across our products and services, including AI technologies, safeguarding both national security and international competitiveness.

Cybersecurity must underpin the federal government’s agenda to ensure secure design, development, deployment, and use of AI systems. Therefore, we recommend the federal government:

- **Advance the use of AI for cyber defense.** Microsoft’s significant investment in AI innovation is aimed at providing cybersecurity defenders with an asymmetric advantage over attackers in the realm of defense. Recent advancements in AI can help address long-standing challenges faced by cyber defenders including talent shortage, burnout, and the scale, speed and efficiency required to defend against cyber-attacks. The federal government should adopt AI capabilities to strengthen its cyber defense and ensure the removal of compliance barriers for piloting modern technology in federal government environments.
- **Leverage existing risk-based cybersecurity standards.** Cybersecurity frameworks, standards, and best practices (“standards”) offer private and public stakeholders a way to build trust and provide assurance. We encourage the federal government to avoid developing duplicative policies by leveraging existing risk-based and outcomes focused cybersecurity standards (e.g., NIST CSF 2.0, SSDF v1.1). Unlike prescriptive cybersecurity requirements, risk-based standards offer flexibility and agility to streamline regulations, drive consistency, and incentivize innovation and cross-border commerce. Developing new standards from scratch is a resource-intensive process and the federal government can save substantial resources using and building on existing risk-based standards while still ensuring robust cybersecurity measures are in place.
- **Streamline regulation.** The current U.S. approach to cybersecurity regulation is fragmented across topics and industry sectors resulting in conflicting compliance regimes that can steer resources from cybersecurity to compliance. We encourage the federal government to streamline AI regulations, as many traditional cybersecurity practices protecting software and hardware—already covered by existing regulations—are applicable to AI.
- **Focus on emerging threats.** Recent advancements in AI may introduce new cybersecurity attack techniques. It will be critical for the federal government, in close collaboration with industry, to focus on addressing emerging AI threats, in particular threats from adversaries and threats targeting critical infrastructure. Additionally, the United States should collaborate with allies to continue to develop and implement international cybersecurity norms to hold accountable those responsible for malicious cyber-attacks. Furthermore, naming attacks by nation-state actors like China, Iran, North Korea, and Russia in venues like the United Nations Security Council is important.
- **Prepare for and address resiliency of U.S. agencies and critical infrastructure to defend against AI-based attacks.** Adversaries may use AI in new ways to help them design and plan sophisticated multipoint attacks, e.g., sequences of infrastructural disruptions across several distinct sectors. It is important to recognize and address new forms of AI-supported attacks by, for example, employing AI red-teaming

studies and using results to guide hardening and introduction of new resiliencies. For additional ideas, see [Applications for artificial intelligence in Department of Defense cyber missions - Microsoft On the Issues](#)

The federal government's leadership in the adoption of AI is crucial for setting a standard across various sectors and ensuring that the United States remains at the forefront of technological innovation. By taking proactive measures in integrating AI into its operations, leveraging pre-existing AI enabled applications, the federal government can not only accelerate the delivery of critical public services but also drive widespread adoption, improve government services for people, and increase confidence for other industries. Furthermore, by leveraging refined AI models, optimized for key use cases, organizations within government can accelerate application development, enhance security through real-time threat detection, ensure compliance, and enable scalable automation. The federal government should embrace the vast array of models available in the AI ecosystem. This leadership is essential for fostering public trust in AI technologies and promoting AI deployment.

Investing in Scientific Research and AI Standards

The United States has a long and successful history in utilizing public-private partnerships for global scientific leadership. Today, America leads the world's innovation economy because of decisions made decades ago to provide publicly funded research infrastructure and financial support to talented scientists and entrepreneurs. Through federal, state, and local governments, investments were made in regional economies and programs that were big bets on American innovation. The great incubators of the 20th century—from Bell Labs to the network of federal national labs—were intentional decisions to bring industry, government, and academia together to accelerate science.

Today, the significance of deploying the same playbook for AI and quantum technologies cannot be overstated. Investments in these fields are essential to pioneering the development of cutting-edge technology solutions that address complex global challenges. To maintain America's role as the global scientific leader, it is crucial to increase federal investment in key areas of AI Research and Development (R&D), support for the National Institute for Standards and Technology (NIST), the Office of the National Coordinator for Health Information Technology (ONC), and the establishment of the National AI Research Resource (NAIRR).

Increase Investment in AI Research and Development

Investing in AI R&D is essential to advance the development of AI models and other innovative AI technology, including enhancing our own understanding of generative AI, driving model innovation, and extending overall human capabilities. Although the CHIPS and Science Act appropriated over \$52 billion for semiconductor technology manufacturing, it only authorized, but did not fund, the 'science' portion of the bill, leaving

an estimated \$50 billion shortfall in aggregate investments. To outcompete a nation like China, which has significantly boosted investment in R&D, the U.S. must accelerate strategic investments in scientific research for future technologies. Experts predict that China will continue to invest substantial resources into next-generation technologies such as AI, advanced manufacturing, clean energy, quantum computing, and semiconductors over the next decade. Therefore, we recommend the federal government:

- **Increase AI R&D spending.** Encourage Congress to increase spending on AI Research and Development (AI R&D). By increasing AI R&D, the United States will not only advance the development of AI but can also properly leverage commercial AI capabilities and increase adoption of AI for Science by empowering our scientists and researchers with the resources they need to accelerate discovery.

Establish the National AI Research Resource (NAIRR)

To secure U.S. global leadership, it is also essential to empower institutions with access to data, computational power, and tools necessary for pioneering AI research and development. Establishing the NAIRR, as recommended by the NAIRR Task Force and outlined in the National Artificial Intelligence Initiative Act of 2020, will democratize access to advanced computational resources and data. This will ensure a diverse range of perspectives in AI research, facilitate the exploration of scientific challenges, and develop the AI workforce of the future. Microsoft is a lead supporter of the NAIRR pilot and looks forward to assisting with guidance and direction in standing up the full NAIRR. Therefore, we recommend the federal government:

- **Support the CREATE AI Act.** Encourage Congress to pass the bipartisan Creating Resources for Every American to Experiment with Artificial Intelligence Act (CREATE AI Act), which would fully establish and fund the NAIRR.

Invest in the National Institute for Standards and Technology

NIST holds a unique role in this AI moment. NIST special publications reflect consultation and engagement with varied stakeholders, offering American companies specific yet flexible guidance in developing frameworks they can use to build and deploy reliable, high-performing AI technologies. NIST thus provides American companies concrete operational tools that facilitate domestic adoption and continuous innovation. At the same time, the quality of NIST guidance has been recognized worldwide. Programs within NIST have tackled some of the most complex technical questions and enabled the United States to maintain a leadership position in multijurisdictional technical dialogues. NIST has also played a critical role in informing the global AI governance conversation, bringing U.S. perspectives to multilateral fora like the Organization for Economic Cooperation and Development (OECD) and the Quadrilateral Security Dialogue, as well as to international standardization bodies like ISO/IEC, where technical experts are actively developing consensus-based, voluntary AI standards. By reinforcing and investing in NIST, the United States can maintain its position in international scientific conversations and inform global

AI governance approaches that underpin growth for American AI. Therefore, we recommend the federal government:

- **Properly staff and fund NIST.** Properly resourcing NIST is vital to ensuring the United States continues to lead in developing the measurement backed science and standards that will facilitate domestic adoption and reinforce the American AI advantage.

Promote the Development of Open, Industry-led Technical Standards

Maintaining a strong role in establishing different forms of soft law is critical for the United States to win the AI race. Standards, open-source software, and codes of conduct are various forms of soft law that provide trust and accountability, which in turn provides AI users with the confidence to adopt and rely on the technology. Standards are built on public-private partnerships that have enabled U.S. technology leadership for more than 50 years.

The U.S. standardization system is based upon foundational principles - industry-led, voluntary, open to all interested stakeholders, and rules-based – which allows market coherence ahead of uniformity when developing standards. This approach has been critical in leading U.S. technology innovations for global market adoption. At no point has this standardization system functioned successfully in a closed or exclusionary capacity.

Using standards and other soft law mechanisms to establish trusted implementations for efficient contracting are reasonable steps to protect users of the technology and to provide a counterweight to industrial policy requirements from other nations. Therefore, we recommend the federal government:

- **Promote the development of open, industry-led technical standards**
Maintaining a strong role in establishing different forms of soft law is critical for the United States to win the AI race. Standards, open-source software, academic papers, and codes of conduct provide trust and accountability, enabling U.S. technology leadership for more than 50 years.

Support Global Engagement to Enhance American AI Leadership

Even with the tremendous technological advancement we have witnessed in AI over the past several years, President Trump's 2019 Executive Order's policy on scientific research remains valuable. As stated in the EO, "The United States must drive technological breakthroughs in AI across the federal government, industry, and academia in order to promote scientific discovery, economic competitiveness, and national security." We could not agree more, yet we have seen that people and organizations will only integrate and build upon breakthrough capabilities and products that are trustworthy and reliable. Trust in technology has underpinned every durable wave of innovation.

There are two components to the trust that supports broad adoption and continuous innovation necessary for American technologies to thrive: first, trust in how technology itself performs; and second, trust that people and organizations can deploy it successfully. To create the conditions for innovation and AI adoption at scale, we need to keep investing in efforts to reinforce trust. Investing in scientific research and voluntary standards supports the first effort. Targeted guardrails can support the second. Having the United States play a leading role in shaping what those guardrails look like and how they can be implemented will create a more hospitable ecosystem both within the United States and across the world for American AI companies of all sizes. To that end, we recommend the federal government:

- **Engage in international governance efforts, such as the G7 Hiroshima AI Process.** The United States should remain actively involved in conversations with our allies to ensure they proceed in a way that reinforces rather than restricts opportunities for American AI by helping to deepen, align, and streamline shared expectations. The Hiroshima AI Process, and its associated reporting framework, is one such work product. Strong U.S. involvement prevents the development of a leadership vacuum that will be filled by countries like China.
- **Build on U.S. AI research leadership via global AI institutes.** The United States should remain involved in conversations with other AI safety and security institutes around the world to ensure American values are represented and integrated into the development of international AI standards and enabling research. This collaboration will also help foster innovation by safeguarding against misuse of AI technologies.
- **Promote AI innovation and deployment in U.S. trade agreements.** Use digital trade policy to promote alignment of regulatory frameworks and rules that support innovation and deployment. These include rules to support cross-border data flows, limit localization of data, protect against forced transfer of technology as a condition for market access, and ensure non-discriminatory treatment of U.S. firms in foreign markets.
- **Promote U.S. friendly AI norms and oppose adversarial narratives in international organizations.** The United States should continue to lead conversations on AI with like-minded partners at the OECD (where President Trump helped deliver the first-ever OECD AI Principles in 2019) and cultivate sensible, action-oriented policies that spur innovation, promote interoperability, and create markets for American business. The United States should similarly engage at the United Nations (UN) by pushing to include the views of American tech companies in UN processes, better shaping UN governance, avoiding politicization, and rejecting Chinese state-centric approaches.

Microsoft appreciates the opportunity to respond to the White House Office of Science and Technology Policy (OSTP) in its request for information for the AI Action Plan. We believe the federal government is uniquely positioned to drive AI adoption and foster the American AI ecosystem through strategic investments, policy development, and international collaboration. By leveraging the principles of smart public-private partnership, science-

based standards, and global engagement, the United States can ensure that its AI technologies supercharge economic competitiveness and national security.

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